

ECE330 – Fall 2023
Exam 1
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90 min

Thursday, September 28, 2023

NAME _____

NetID _____

Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- To get credit, you must SHOW ALL your work and provide explanations where needed.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 10 pages (including this one) with 4 problems.

Problem 1 (25 Points)

Note 1: Parts **a)**, **b)** and **c)** all refer to the same system described below.

Note 2: You may leave complex numbers in either rectangular or polar form.

A single-phase source with a voltage $v(t) = 157 \cos(377t + 15^\circ)V$ is serving two loads connected in parallel:

- Load 1 consumes 800 W of real power and 400 VARs of reactive power.
- Load 2 is an impedance of magnitude 14 Ohms and angle 20 degrees.

a) Compute the time-domain currents for Load 1 and Load 2.

$$i_1(t) = \underline{11.394 \cos(377t - 11.565^\circ) \text{ A}}$$

$$i_2(t) = \underline{11.215 \cos(377t - 5^\circ) \text{ A}}$$

$$\vec{V} = \frac{157}{\sqrt{2}} \angle 15^\circ = 111.016 \angle 15^\circ \text{ V}$$

Load 1: $\vec{S}_1 = 800 + j400 \text{ VA}$

$$\vec{S}_1 = \vec{V} \cdot \vec{I}_1^* \Rightarrow \vec{I}_1 = \left(\frac{\vec{S}_1}{\vec{V}_1} \right)^*$$

$$\vec{I}_1 = \frac{800 - j400}{111.016 \angle -15^\circ} = 8.0567 \angle -11.565^\circ \text{ A}$$

$$I_1(t) = 11.394 \cos(377t - 11.565^\circ)$$

Load 2: $\vec{Z} = 14 \angle 20^\circ \Omega$

$$\vec{I}_2 = \frac{\vec{V}_2}{\vec{Z}} = \frac{111.016 \angle 15^\circ}{14 \angle 20^\circ} = 7.93 \angle -5^\circ \text{ A}$$

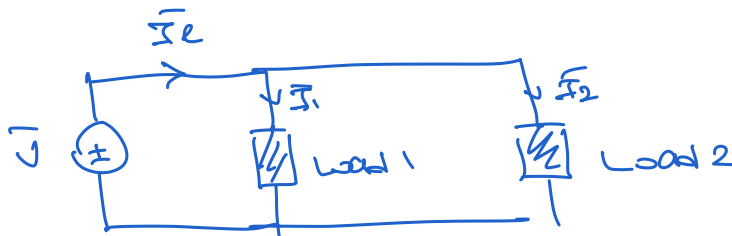
$$I_2(t) = 11.215 \cos(377t - 5^\circ) \text{ A}$$

b) Find the total complex power supplied by the source.

$$\overline{S}_{tot} = \underline{1627.207 + j 701.089 \text{ VA}}$$

$$\vec{I}_L = \vec{I}_1 + \vec{I}_2 = 15.96 \angle -8.309^\circ \text{ A}$$

$$\begin{aligned} \overline{S}_{tot} &= \vec{V} \cdot \vec{I}_L^* = (111.016 \angle 15^\circ) (15.96 \angle 8.309^\circ) \\ &= 1627.207 + j 701.089 \text{ VA} \\ &= 1771.815 \angle 23.309^\circ \text{ VA} \end{aligned}$$



c) Find the reactance of the capacitor (in Ohms) that you could add as a third load in parallel to achieve PF of unity.

$$X_C = \underline{-17.579 \, \Omega}$$

$$Q_{tot} = 701.089 \, \text{VAR}$$

$$Q_{new} = Q_{tot} + Q_C = 0 \Rightarrow Q_C = -701.089 \, \text{VAR}$$

$$Q_C = \frac{V^2}{X_C} \Rightarrow X_C = \frac{V^2}{Q_C} = \frac{157^2}{2 Q_C} = -17.579 \, \Omega$$

Problem 2 (25 Points)

Note: Parts a), b) and c) all refer to the same system described below.

Two loads are supplied by a three-phase wye-connected source with line voltage 11kV:

1. 300 kilowatts (3-phase), wye connected, PF = 0.8 (lagging).
2. 3 Amps (phase current), delta connected, PF = 0 (leading).

a) Find the total real power and total reactive power of the two loads.

$$P_{\text{tot}} = \underline{300 \text{ kW}}$$

$$Q_{\text{tot}} = \underline{126 \text{ kVAR}}$$

Load 1:

$$P_1 = 300 \text{ kW}$$

$$Q_1 = 300 \tan(\cos^{-1} 0.8) = 225 \text{ kVAR}$$

Load 2

$$V_L = 11 \text{ kV}$$

$$I_L = 3\sqrt{3} \text{ A}$$

$$S_2 = \sqrt{3} V_L I_L = 99 \text{ kVA}$$

$$\text{PF} = 0 \text{ leading} \quad Q_2 = -99 \text{ kVAR}$$

$$P_{\text{tot}} = P_1 + \cancel{P_2} \quad 0$$

$$Q_{\text{tot}} = Q_1 + Q_2$$

b) Compute the total line current magnitude being supplied by the source.

$$I_L = \underline{17.078 \text{ A}}$$

$$S_{\text{tot}} = \sqrt{P_{\text{tot}}^2 + Q_{\text{tot}}^2} = 325.386 \text{ kVA}$$

$$S_{\text{tot}} = \sqrt{3} V_L \cdot I_{\text{tot}} \Rightarrow I_{\text{tot}} = 17.078 \text{ A}$$

c) What is the total line current magnitude if a sufficiently large amount of capacitance is added to make the overall power factor unity?

$$I_{L,\text{new}} = \underline{15.746 \text{ A}}$$

$$Q_{\text{new}} = 0$$

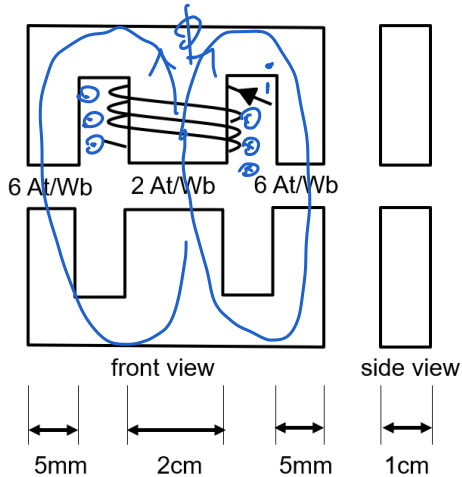
$$S_{\text{tot}} = P_{\text{tot}} = 300 \text{ kVA}$$

$$S_{\text{tot}} = \sqrt{3} V_L \cdot I_{\text{new}} \Rightarrow I_{\text{new}} = 15.746 \text{ A}$$

Problem 3 (25 Points)

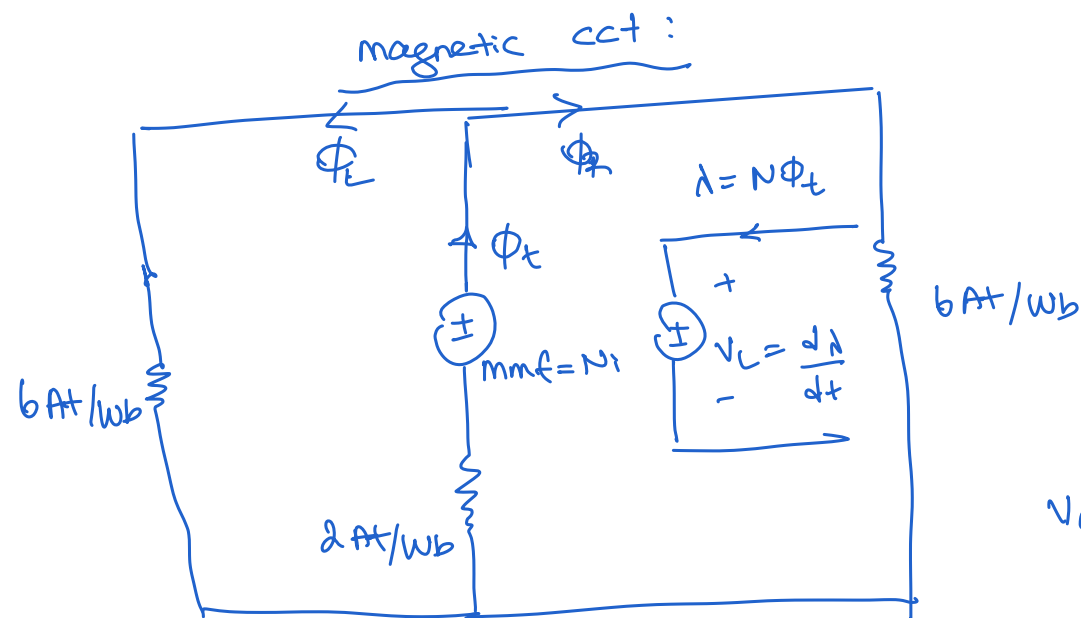
Note: Parts a), b) and c) all refer to the same structure described below.

Consider the iron structure shown. Assume that $\mu = \infty$ in the iron. The coil has $N = 12$ turns.



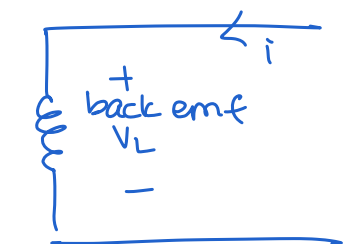
a) Draw the magnetic circuit for the structure. Please take care to orient the polarities correctly. Label the mmfs. Use the magnetic circuit to compute the inductance of the structure.

$$L = \underline{144/5 \text{ H}}$$



$$R_{eq} = (6 // 6) + 2 = 5 \text{ At/Wb}$$

Electric cct:



$$V_L = L \frac{di}{dt}$$

$$V_L = L \frac{di}{dt} = \frac{d\lambda}{dt}$$

$$L = \frac{d\lambda}{di} = N \cdot \frac{d\Phi}{di}$$

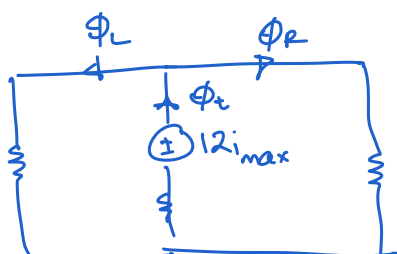
$$= N \frac{d}{di} \left(\frac{Ni}{R_{eq}} \right)$$

$$L = \frac{144}{5} \text{ H}$$

$$L = N^2 / R_{eq}$$

b) The flux density in each of three legs must be kept at equal or below $B = 0.1 \text{ T}$, or else the core will saturate. What is the maximum current $i_{\max}$ that we can drive the coil without causing saturation?

$$i_{\max} = \underline{4.167 \mu\text{A}}$$



Magnetic flux density of the middle leg

$$\Phi_t = \Phi_L + \Phi_R = 2\Phi_L \quad (\Phi_L = \Phi_R)$$

$$B_t = \frac{2\Phi_L}{A_t} = \frac{2\Phi_L}{2 \text{ cm} \times 1 \text{ cm}}$$

$$B_L = \frac{\Phi_L}{A_L} = \frac{\Phi_L}{0.5 \text{ cm} \times 1 \text{ cm}}$$

$B_L = 2B_t \Rightarrow$ Bottleneck is the left/right legs.

$$\Phi_L = \left(\frac{Ni}{R_{eq}} \right) / 2 = \frac{bi}{5}$$

$$B_L = \frac{bi}{5 (5 \times 10^{-5} \text{ m}^2)} = 24000i \Rightarrow 24000i \leq 0.1 \text{ T}$$

$$i_{\max} = 4.167 \mu\text{A}$$

c) The reluctances stated in the diagram were computed neglecting fringing. If the effects of fringing were instead accounted for, how would your answer to part b) change? Would the maximum current $i_{\max}$ without causing saturation increase or decrease? Explain.

Note: Credit will only be awarded for the explanation. No credit will be awarded for the correct choice.

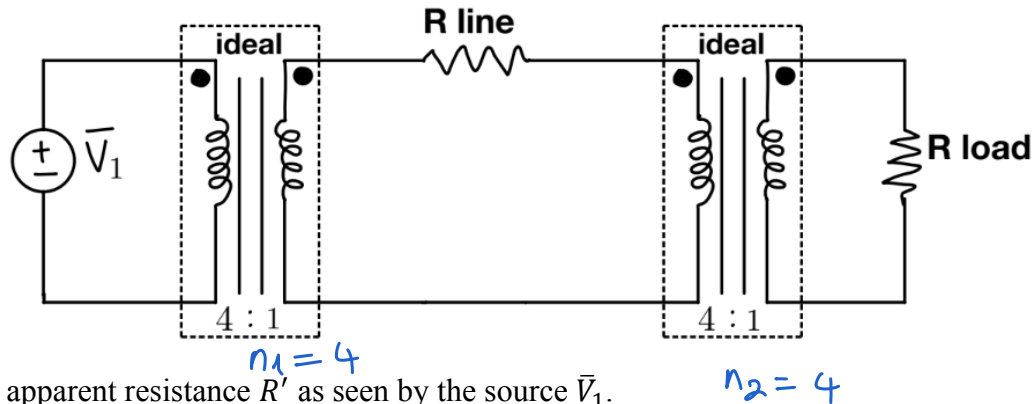
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Explanation:

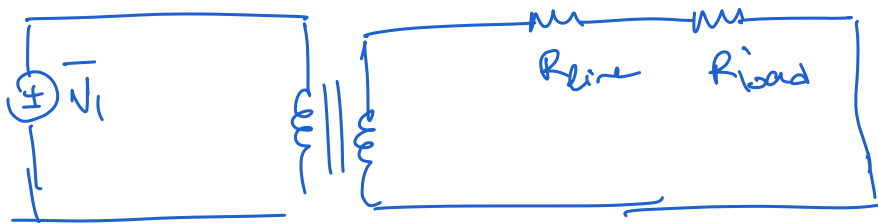
$$R_{\text{new}} = \frac{1}{\mu_0} \frac{\text{gap}}{(\text{gap} + \text{width})(\text{gap} + \text{depth})} \Rightarrow R_{\text{new}} < R_{eq}$$

$$\Phi_R = Ni_{\max} \Rightarrow \frac{B_{\max}}{A} \cdot A \downarrow = Ni_{\max} \downarrow$$

$$i_{\max, \text{new}} < i_{\max} \quad \text{since } R_{\text{new}} < R_{eq}$$

Problem 4 (25 Points)Note: Parts a) and b) are unrelated.a) The circuit shown below has $R_{load} = 2 \Omega$ and $R_{line} = 10 \Omega$:Find the apparent resistance R' as seen by the source $\bar{V}_1$.

$$R' = \underline{672 \Omega}$$

 $\Rightarrow$ Refer R_{load} to the primary side:

$$n_2 = \frac{4}{1}$$

$$R'_{load} = n_2^2 R_{load}$$

$$= 4^2 \cdot 2$$

$$= 32 \Omega$$

 $\Rightarrow$ Refer $(R_{line} + R'_{load})$ to the primary side

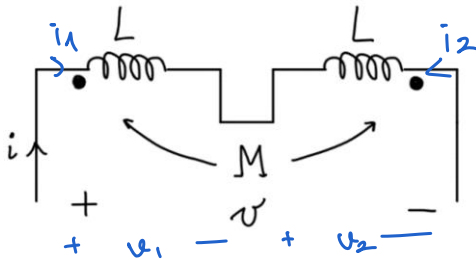
$$n_1 = 4$$

$$R' = n_1^2 (R_{tot}) = 4^2 \times 42 = 672 \Omega$$

$$= 32 + R_{line}$$

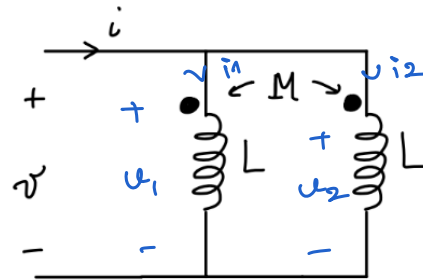
b) Two identical but mutually coupled coils are measured in two ways:

In series with the undotted terminal of one coil connected to the dotted terminal of the other coil.



This series structure has equivalent inductance of $L_{eq} = 8 \text{ mH}$.

In parallel with the dotted terminals connected together and the undotted terminals connected together.



This parallel structure has equivalent inductance of $L_{eq} = 5 \text{ mH}$.

Find the self-inductance of each coil L and the mutual inductance between two coils M .

$$L = \underline{7 \text{ mH}} \quad M = \underline{3 \text{ mH}}$$

Series cct:

$$i_1 = i, \quad i_2 = -i, \quad v = v_1 - v_2$$

$$\begin{aligned} v_1 &= L \frac{di_1}{dt} + M \frac{di_2}{dt} \Rightarrow v_1 = (L - M) \frac{di}{dt} \\ v_2 &= M \frac{di_1}{dt} + L \frac{di_2}{dt} \Rightarrow v_2 = (M - L) \frac{di}{dt} \end{aligned}$$

$$\begin{aligned} v &= v_1 - v_2 = (2L - 2M) \frac{di}{dt} \\ &= L_{eq} \frac{di}{dt}, \quad L_{eq} = 8 \text{ mH} \end{aligned}$$

$$L - M = 4 \text{ mH}$$

Parallel cct:

$$i = i_1 + i_2, \quad i_1 = i_2, \quad v = v_1 = v_2$$

$$v_1 = v_2 = v$$

$$L \frac{di_1}{dt} + M \frac{di_2}{dt} = M \frac{di_1}{dt} + L \frac{di_2}{dt}$$

$$(L - M) \frac{di_1}{dt} = (L - M) \frac{di_2}{dt} \Rightarrow i_1 = i_2$$

$$i = 2i_1$$

$$v_1 = (L + M) \frac{di_1}{dt}, \quad v_1 = v = L_{eq} \frac{d(2i_1)}{dt} \Rightarrow v_1 = (10 \times 10^{-3}) \frac{di_1}{dt} \Rightarrow M + L = 10 \text{ mH}$$

$$\begin{aligned} L &= 7 \text{ mH} \\ M &= 3 \text{ mH} \end{aligned}$$